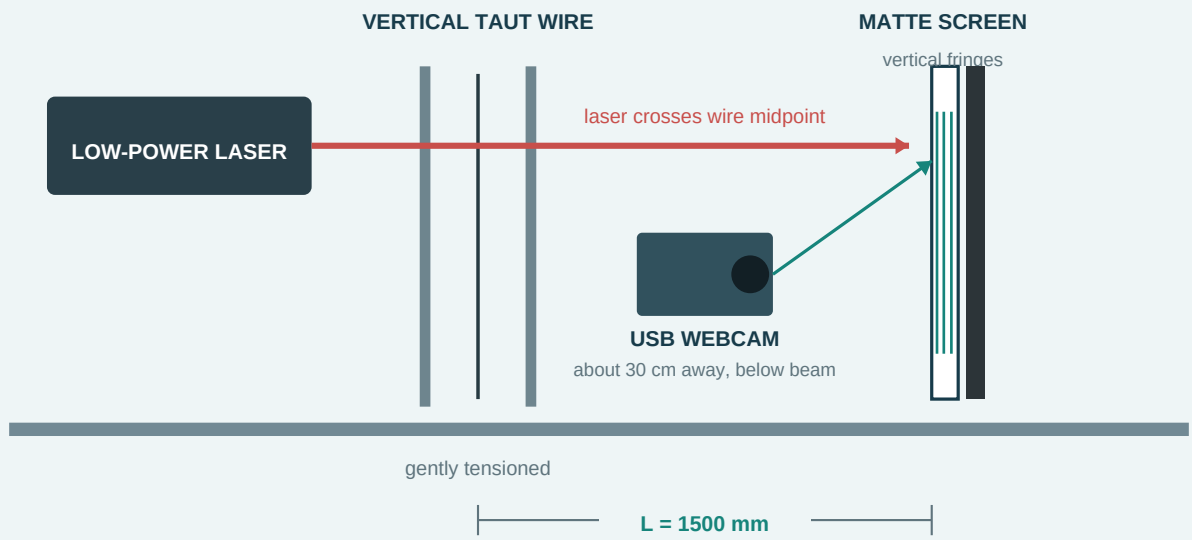
FINAL MODEL

Automatic optical wire micrometer

Borrowed college optical bench | Purchased laser | DIY riders

A laser passes a vertically held thin wire. The wire creates bright and dark diffraction bands on a matte screen. A USB webcam films the screen; laptop software finds the first dark bands and calculates the unknown diameter.

PHYSICAL BUILD - SIDE VIEW (NOT TO SCALE)



Signal path

1 Diffraction
wire creates a central bright band and dark minima

→

2 Webcam
captures the pattern on the matte card

→

3 Laptop
OpenCV extracts a brightness profile and minima

→

4 Result
diameter appears live; S saves a CSV row

No OV7670 or Pico camera bus: the laptop processes the USB webcam directly.

Use a matte card and place the camera off the beam axis. Keep a black beam stop behind the screen. Do not remove the webcam lens or aim the laser into the camera.

BUILD PLAN

Budget, DIY riders and assembly

Borrow the optical bench; purchase the laser; make riders to fit the bench.

Cost plan - fill in prices from your actual purchases

Item	Plan	Cost
Optical bench	Borrow from college	Rs 0
Red laser	Purchase a labelled low-power unit; note its wavelength	Actual price
Riders and holders	Make/adapt to fit the borrowed bench; keep mounts rigid	DIY cost
Webcam and laptop	Use the selected webcam and available laptop	Actual cost
Screen, stop, sample	Make a matte screen holder and beam stop; source the sample	DIY cost

Budget check: the old total excluded a purchased laser and is withdrawn. Add the laser, webcam and DIY rider costs from actual prices, then check the combined total against the Rs 1,500 ceiling before buying. Choose a laser with a trustworthy safety/class and wavelength label; cheap unlabelled modules may not meet the safety requirement.

Assembly and alignment

1. **Check the borrowed bench**

   Measure the rail profile and rider clamp dimensions first. Make or adapt the riders to slide and lock securely on that exact bench.

2. **Mount laser and wire**

   Fix the purchased laser in a rigid DIY rider. Make a vertical wire frame with upper and lower clips; stretch the sample gently, with clips outside the beam.

3. **Set the optical distance**

   Set the wire-to-screen distance to 1500 mm. Keep the screen perpendicular to the beam and put a nonreflective beam stop behind it.

4. **Aim and frame**

   Aim the beam through the wire midpoint. Put the webcam about 30 cm before the screen, below the beam, focused on the card. Keep the live pattern centered.

5. **Check the geometry**

   The wire is vertical, the beam is horizontal, and the nearest dark minima should appear on both sides of the central bright band. Shade out room light.

Position guide: Rider 1 holds the laser; Rider 2 holds the DIY wire frame; Rider 3 supports the screen. Keep the screen 1500 mm from the wire. The camera is separately supported about 30 cm in front of the screen and below the beam.

MEASUREMENT

Calibration, calculation and presentation

The unknown sample is measured automatically after a one-time reference capture for each run.

Run the software

Install Python 3.10 or later, then install the libraries:

```
python -m pip install opencv-python numpy
```

Connect the USB webcam, close video-call apps, and start the program. Substitute the wavelength printed on the laser label; 650 nm below is an example.

```
python laser_micrometer.py --camera 0 --distance-mm 1500 --wavelength-nm 650 --reference-diameter-um 127
```

Fresh reference calibration - recommended

1. Place a reference wire with a known diameter in the holder. Its diameter should be measured with a screw gauge or supplied by a trustworthy specification.
2. Wait until the live image shows a stable central bright band and a pair of dark minima. Press **R**. The app stores the reference span for this run.
3. Replace the reference with the unknown wire without moving the laser, camera, screen, or holder. The live diameter updates automatically.
4. Press **S** to save a valid reading to `measurements.csv`; press **Q** to quit. The app asks for a fresh calibration next time it starts.

Physics relation

d = 2 lambda L / Delta-x

d = wire diameter | lambda = laser wavelength | L = wire-to-screen distance

Delta-x = separation between first dark minima on the two sides of the centre

Reference method: d_unknown = d_reference x span_reference / span_unknown

For the project demonstration

Show the live webcam image, marked minima, brightness profile, calculated diameter, and saved CSV result. Explain that the wire diffraction pattern is equivalent to the pattern from a slit of the same width (Babinet principle). Compare the reference result with the screw-gauge value.

Safety: use a labelled teacher-approved low-power laser; never look into the beam or point it at anyone.